# Full PI-VSG + VC-ICL — Nonlinear DAE Model

*Matas-Díaz et al., JMPSCE 2025, §II.A–II.C.1, eqs. (1)–(16)*

## System architecture

**$dx/dt = f(x, y, u)$** (13 differential states) **$0 = g(x, y, u)$** (8 algebraic variables)

LCL filter + Grid Thevenin ➔ PI-VSG OCL (APCL + RPCL) ➔ VC-ICL (outer: voltage loop / inner: current loop)

## 1. State variables x (13)

| | | |
|---|---|---|
| $i_{td}, i_{tq}$ | VSC-side inductor currents, dq | eq. 1 |
| $v_{md}, v_{mq}$ | Capacitor voltage, dq | eq. 2a |
| $i_{sd}, i_{sq}$ | Grid-side currents, dq | eqs. 2b+4 |
| $\psi$ | Phase angle difference | eq. 11a |
| $\xi_p$ | Integral of active power error | eq. 11b |
| $\xi_q$ | Integral of reactive power error | eq. 9b |
| $\xi_{vd}, \xi_{vq}$ | Voltage error integrals, dq — outer loop | eq. 14 |
| $\xi_{id}, \xi_{iq}$ | Current error integrals, dq — inner loop | eq. 15 |

## 2. Algebraic variables y (8)

| | | |
|---|---|---|
| $v_{td}, v_{tq}$ | VSC terminal voltage (current-loop output) | eq. 15 |
| $\omega_{vsg}$ | VSG angular speed (PI-VSG) | eq. 11a |
| $E$ | Virtual EMF amplitude (RPCL) | eq. 9a |
| $v_{sd}, v_{sq}$ | PCC voltage (grid Thevenin + filter) | eqs. 2b+4 |
| $p, q$ | Active/reactive power at POI | POI |

## 3. Intermediate expressions (substituted inline — not states)

| | | |
|---|---|---|
| $v^*_{md}, v^*_{mq}$ | Capacitor voltage references (virtual impedance) | eq. 13 |
| $i^*_{td}, i^*_{tq}$ | VSC-side current references (PI voltage loop) | eq. 14 |

## 4. External inputs u

| | |
|---|---|
| $p^*, q^*$ | Active/reactive power references |
| $v_{gd}, v_{gq}$ | Grid Thevenin voltage, dq |

## 5. Parameters (default values, Table I)

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| $L_t$ | 1.25 mH | H | VSC-side inductor | $R_t$ | 0.04 | Ω | VSC-side resistance |
| $C_f$ | 4 μF | F | Filter capacitor | $R_d$ | 10.0 | Ω | Parallel damping |
| $L_s$ | 1.25 mH | H | Grid-side inductor | $R_s$ | 0.04 | Ω | Grid-side resistance |
| $L_g$ | 2.0 mH | H | Grid Thevenin inductor | $R_g$ | 2.0 | Ω | Grid Thevenin resistance |
| $k_{pp}$ | 0.0012 | rad/s/W | PI-VSG proportional gain | $k_{ip}$ | 0.0016 | rad/s²/W | PI-VSG integral gain |
| $k_{pq}$ | 0.0016 | V/W | RPCL proportional gain | $k_{iq}$ | 0.0163 | V/W/s | RPCL integral gain |
| $E_0$ | 230.0 | V | Rated virtual EMF | $\omega_0$ | 100π | rad/s | Rated speed (50 Hz) |
| $R_v$ | 0.0 | Ω | Virtual resistance | $X_v$ | 0.08 | Ω | Virtual reactance |
| $k^{vc}_{pv}$ | 0.1 | A/V | Voltage-loop P gain | $k^{vc}_{iv}$ | 0.1 | A/V/s | Voltage-loop I gain |
| $k^{vc}_{pi}$ | 0.1 | V/A | Current-loop P gain | $k^{vc}_{ii}$ | 15.1 | V/A/s | Current-loop I gain |

## Full PI-VSG + VC-ICL — Differential and Algebraic Equations

### A. LCL Filter (eqs. 1, 2a, 2b+4) — 6 states

VSC-side inductor:

$$\dot{i}_{td} = (v_{td} - R_t i_{td} + \omega_{vsg} L_t i_{tq} - v_{md}) \,/\, L_t$$

$$\dot{i}_{tq} = (v_{tq} - R_t i_{tq} - \omega_{vsg} L_t i_{td} - v_{mq}) \,/\, L_t$$

Capacitor (parallel R_d damping):

$$\dot{v}_{md} = (i_{td} - i_{sd} - v_{md}/R_d + \omega_{vsg} C_f v_{mq}) \,/\, C_f$$

$$\dot{v}_{mq} = (i_{tq} - i_{sq} - v_{mq}/R_d - \omega_{vsg} C_f v_{md}) \,/\, C_f$$

Grid-side inductor + grid Thevenin combined:

$$\dot{i}_{sd} = (v_{md} - v_{gd} - (R_s{+}R_g) i_{sd} + \omega_{vsg}(L_s{+}L_g) i_{sq}) \,/\, (L_s{+}L_g)$$

$$\dot{i}_{sq} = (v_{mq} - v_{gq} - (R_s{+}R_g) i_{sq} - \omega_{vsg}(L_s{+}L_g) i_{sd}) \,/\, (L_s{+}L_g)$$

### B. PI-VSG Outer Control Loop (eqs. 9, 11) — 3 states

APCL (eq. 11):

$$\dot{\psi} = -k_{pp}(p^* - p) - k_{ip}\xi_p$$

$$\dot{\xi}_p = p^* - p$$

RPCL (eq. 9b):

$$\dot{\xi}_q = q^* - q$$

### C. VC-ICL — Cascaded PI Controllers (eqs. 13–15) — 4 states

Virtual impedance (eq. 13, $e^T=[0,E]^T$, $Z_v=[[R_v,-X_v],[X_v,R_v]]$) → substituted inline:

$$v_{md}^* = -R_v i_{sd} + X_v i_{sq}$$

$$v_{mq}^* = E - X_v i_{sd} - R_v i_{sq}$$

Outer voltage loop (eq. 14): voltage error integrals + current references:

$$\dot{\xi}_{vd} = v_{md}^* - v_{md}$$

$$\dot{\xi}_{vq} = v_{mq}^* - v_{mq}$$

$$i_{td}^* = k_{pv}^{vc}(v_{md}^* - v_{md}) + k_{iv}^{vc}\xi_{vd} - \omega_{vsg} C_f v_{mq} + i_{sd}$$

$$i_{tq}^* = k_{pv}^{vc}(v_{mq}^* - v_{mq}) + k_{iv}^{vc}\xi_{vq} + \omega_{vsg} C_f v_{md} + i_{sq}$$

Inner current loop (eq. 15): current error integrals:

$$\dot{\xi}_{id} = i_{td}^* - i_{td}$$

$$\dot{\xi}_{iq} = i_{tq}^* - i_{tq}$$

### D. Algebraic equations 0 = g(x, y, u) — 8 equations

Inner current-loop terminal voltage (eq. 15):

$$0 = v_{td} - \left[k_{pi}^{vc}(i_{td}^* - i_{td}) + k_{ii}^{vc}\xi_{id} + \omega_{vsg} L_t i_{tq} + v_{md}\right]$$

$$0 = v_{tq} - \left[k_{pi}^{vc}(i_{tq}^* - i_{tq}) + k_{ii}^{vc}\xi_{iq} - \omega_{vsg} L_t i_{td} + v_{mq}\right]$$

PI-VSG speed and EMF:

$$0 = \omega_{vsg} - \omega_0 - k_{pp}(p^* - p) - k_{ip}\xi_p$$

$$0 = E - E_0 - k_{pq}(q^* - q) - k_{iq}\xi_q$$

PCC voltage ($\alpha = L_g/(L_s+L_g)$, $R_{mix} = (R_g \cdot L_s - R_s \cdot L_g)/(L_s+L_g)$):

$$0 = v_{sd} - [(1-\alpha)v_{gd} + \alpha v_{md} + R_{mix}\, i_{sd}]$$

$$0 = v_{sq} - [(1-\alpha)v_{gq} + \alpha v_{mq} + R_{mix}\, i_{sq}]$$

Active and reactive power at POI:

$$0 = p - (v_{sd}\, i_{sd} + v_{sq}\, i_{sq})$$

$$0 = q - (v_{sq}\, i_{sd} - v_{sd}\, i_{sq})$$